DAILY PRACTICE CALENDAR

Precalculus • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This calendar follows the **AP Precalculus** course framework, like the Precalculus course on fangmath.com: Units 1–2 match AP Unit 1, Unit 3 matches AP Unit 2, Units 4–5 match AP Unit 3, and Unit 6 matches AP Unit 4. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Average Rates of Change
2	Aug 23–Aug 27	Rates of Change & Concavity
3	Aug 30–Sep 3	Polynomial Functions & Zeros
4	Sep 6–Sep 10	End Behavior & Polynomial Inequalities
5	Sep 13–Sep 17	Rational Functions & Asymptotes
6	Sep 20–Sep 24	Holes, Zeros & Rational Inequalities
7	Sep 27–Oct 1	Equivalent Representations
8	Oct 4–Oct 8	Transformations of Functions
9	Oct 11–Oct 15	Choosing a Function Model
10	Oct 18–Oct 22	Review & Checkpoint 1
11	Oct 25–Oct 29	Arithmetic & Geometric Sequences
12	Nov 1–Nov 5	Exponential Functions & Models
13	Nov 8–Nov 12	Composition & Inverse Functions
14	Nov 15–Nov 19	Logarithms & Their Properties
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Exponential & Log Equations; Semi-Log Plots
16	Dec 6–Dec 10	Periodic Phenomena & the Unit Circle
17	Dec 13–Dec 17	Sinusoidal Functions
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Sinusoidal Models & the Tangent Function
19	Jan 10–Jan 14	Inverse Trigonometric Functions
20	Jan 17–Jan 21	Mid-Year Review & Checkpoint
21	Jan 24–Jan 28	Trig Equations & Identities
22	Jan 31–Feb 4	Sum & Double-Angle Identities
23	Feb 7–Feb 11	Secant, Cosecant & Cotangent
24	Feb 14–Feb 18	Polar Coordinates & Complex Numbers
25	Feb 21–Feb 25	Polar Graphs
26	Feb 28–Mar 3	Parametric Functions
27	Mar 6–Mar 10	Conic Sections
28	Mar 13–Mar 17	Vectors
29	Mar 20–Mar 24	Matrices
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Matrices as Transformations
31	Apr 10–Apr 14	Matrix Models
32	Apr 17–Apr 21	Year-End Review: Functions & Models
33	Apr 24–Apr 28	Year-End Review & Practice Exam
34	May 1–May 5	Final Checkpoint & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027

Unit 1: Rates of Change & Polynomial Functions

Average Rates of Change

Core: Average rate of change on $[a, b]$ is $\dfrac{f(b) - f(a)}{b - a}$.

Mon Aug 16

From a formula

TRY Find the average rate of change of $f(x) = x^2$ on $[1, 4]$.

Tue Aug 17

From a table

TRY Time t: 0, 2, 5; distance d: 0, 30, 90. Find the average rate of change on $[2, 5]$.

Wed Aug 18

Units matter

TRY A town grew from 1,200 to 1,800 people in 4 years. Give the average rate of change with units.

Thu Aug 19

Comparing intervals

TRY For $f(x) = 2^x$, compare the average rates of change on $[0, 2]$ and $[2, 4]$.

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027

Unit 1: Rates of Change & Polynomial Functions

Rates of Change & Concavity

Core: When the rate of change is increasing, the graph is concave up.

Mon Aug 23

Increasing & decreasing

TRY On what interval is $f(x) = x^2 - 4x$ decreasing?

Tue Aug 24

Concavity from rates

TRY Average rates over equal consecutive intervals are 2, 5, 9, 14. Is the graph concave up or down?

Wed Aug 25

Concavity of a graph

TRY Is $y = \sqrt{x}$ concave up or concave down for $x > 0$?

Thu Aug 26

Points of inflection

TRY Where does $y = x^3$ change concavity?

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027

Unit 1: Rates of Change & Polynomial Functions

Polynomial Functions & Zeros

Core: A degree-n polynomial has n complex zeros, counted with multiplicity.

Mon Aug 30

Zeros & multiplicity

TRY Find the zeros of $f(x) = x^2(x - 3)^3$ and their multiplicities.

Tue Aug 31

Building a polynomial

TRY A cubic with leading coefficient 1 has zeros -1, 2 and 4. Find its constant term.

Wed Sep 1

Complex zeros

TRY What is the least degree of a polynomial with real coefficients and zeros 3 and $2i$?

Thu Sep 2

Local extrema

TRY What is the greatest number of local extrema a degree-5 polynomial can have?

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027

Unit 1: Rates of Change & Polynomial Functions

End Behavior & Polynomial Inequalities

Core: The leading term controls the end behavior; a sign chart solves inequalities.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

End behavior

TRY As $x \to \infty$, what does $-3x^4 + x$ approach?

Tue Sep 7

Even & odd functions

TRY Is $f(x) = x^4 - 2x^2$ even, odd or neither?

Wed Sep 8

Polynomial inequalities

TRY Solve $(x - 1)(x + 2) > 0$.

Thu Sep 9

Sign charts

TRY Solve $x(x - 3)^2 \leq 0$.

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Rational Functions & Modeling

Rational Functions & Asymptotes **Core:** Compare the degrees of the numerator and denominator to find the end behavior.

Mon Sep 13
Vertical asymptotes
TRY Find the vertical asymptotes of $f(x) = \dfrac{x+2}{(x-1)(x+3)}$.

Tue Sep 14
Horizontal asymptotes
TRY Find the horizontal asymptote of $f(x) = \dfrac{2x^2-1}{5x^2+x}$.

Wed Sep 15
Slant asymptotes
TRY Find the slant asymptote of $f(x) = \dfrac{x^2+1}{x-1}$.

Thu Sep 16
End behavior
TRY As $x \to \infty$, what does $\dfrac{x^3}{x^2+1}$ approach?

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Rational Functions & Modeling

Holes, Zeros & Rational Inequalities **Core:** A factor that cancels makes a hole, not an asymptote.

Mon Sep 20
Holes
TRY Where is the hole in the graph of $f(x) = \dfrac{x^2-4}{x-2}$?

Tue Sep 21
Zeros
TRY Find the zeros of $f(x) = \dfrac{x^2-9}{x+3}$.

Wed Sep 22
Rational inequalities
TRY Solve $\dfrac{x-2}{x+1} \geq 0$.

Thu Sep 23
Near an asymptote
TRY What happens to $f(x) = \dfrac{1}{(x-2)^2}$ as $x \to 2$?

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Rational Functions & Modeling

Equivalent Representations **Core:** Divide, expand or factor to reveal a function's key features.

Mon Sep 27
Polynomial division
TRY Simplify $\dfrac{x^3-1}{x-1}$.

Tue Sep 28
Binomial theorem
TRY Expand $(x+2)^4$.

Wed Sep 29
Rewriting rational functions
TRY Write $\dfrac{2x+5}{x+1}$ as $2 + \dfrac{\square}{x+1}$.

Thu Sep 30
Hidden quadratics
TRY Solve $x^4 - 13x^2 + 36 = 0$.

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Rational Functions & Modeling

Transformations of Functions **Core:** $g(x) = a \cdot f(b(x-h)) + k$ stretches, shifts and reflects f.

Mon Oct 4
Shifts & stretches
TRY Describe $g(x) = 2f(x-3) + 1$ in terms of f.

Tue Oct 5
Horizontal dilations
TRY Describe $g(x) = f(2x)$ in terms of f.

Wed Oct 6
Range after a transformation
TRY f has range $[-1, 4]$. Find the range of $3f(x) - 2$.

Thu Oct 7
Reflections
TRY $(4, -2)$ is on the graph of f. What point must be on $g(x) = -f(-x)$?

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Rational Functions & Modeling

Choosing a Function Model

Core: Let the data's pattern of change choose the model.

Mon Oct 11
Linear or quadratic?
TRY Data with equally spaced inputs have constant second differences of 6. Which model fits?

Tue Oct 12
Piecewise models
TRY Shipping costs $5 up to 2 lb, plus $2 for each extra pound. What does a 5 lb package cost?

Wed Oct 13
Fitting a quadratic
TRY Find the quadratic through $(0, 3)$, $(1, 6)$ and $(2, 13)$.

Thu Oct 14
Model limits
TRY Why might a linear model fail for long-term population growth?

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Rational Functions & Modeling

Review & Checkpoint 1

Core: Pull Units 1–2 together and fix any weak spots.

Mon Oct 18
Review: rates of change
TRY Find the average rate of change of $f(x) = x^3$ on $[-1, 2]$.

Tue Oct 19
Review: zeros
TRY Find the zeros of $x^3 - 4x^2 + 4x$.

Wed Oct 20
Review: asymptotes
TRY Find the asymptotes of $f(x) = \dfrac{3x - 1}{x + 4}$.

Thu Oct 21
Review: transformations
TRY Find the vertex of $y = -(x + 1)^2 + 4$.

Fri Oct 22
Checkpoint 1
Units 1–2 test. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Exponential & Logarithmic Functions

Arithmetic & Geometric Sequences

Core: Arithmetic sequences grow like linear functions; geometric sequences grow like exponentials.

Mon Oct 25
Arithmetic sequences
TRY If $a_1 = 7$ and $d = -3$, find a_{10}.

Tue Oct 26
Geometric sequences
TRY If $g_1 = 3$ and $r = 2$, find g_8.

Wed Oct 27
Making the connection
TRY Arithmetic is to linear as geometric is to _____.

Thu Oct 28
Finding the ratio
TRY A geometric sequence has $g_2 = 12$ and $g_5 = 96$. Find r.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Exponential & Logarithmic Functions

Exponential Functions & Models

Core: Equal input steps multiply the output by the same factor.

Mon Nov 1
Growth factors
TRY A quantity doubles every 5 years. What is its yearly growth factor?

Tue Nov 2
Writing models
TRY A value starts at 800 and falls 12% a year. Write a model.

Wed Nov 3
Equivalent forms
TRY Write $3 \cdot 4^{x+1}$ in the form $a \cdot 4^x$.

Thu Nov 4
Continuous growth
TRY If $P(t) = 100e^{0.03t}$, find $P(10)$.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Exponential & Logarithmic Functions

Composition & Inverse Functions

Core: Inverse functions swap inputs and outputs.

Mon Nov 8
Composition
TRY If $f(x) = \sqrt{x}$ and $g(x) = x^2 + 7$, find $f(g(3))$.

Tue Nov 9
Decomposition
TRY Write $h(x) = (3x - 1)^5$ as $f(g(x))$.

Wed Nov 10
Inverses
TRY Find the inverse of $f(x) = 2^x - 1$.

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Domain of an inverse
TRY If $f(x) = e^x + 2$, what is the domain of f^{-1}?

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Exponential & Logarithmic Functions

Logarithms & Their Properties

Core: Logarithms are the inverses of exponential functions.

Mon Nov 15
Evaluating
TRY Find $\log_2 \frac{1}{16}$.

Tue Nov 16
Product rule
TRY Compute $\log 5 + \log 20$.

Wed Nov 17
Expanding
TRY Expand $\ln\left(x^2\sqrt{y}\right)$.

Thu Nov 18
Log graphs
TRY Find the vertical asymptote of $y = \ln(x + 3)$.

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Exponential & Logarithmic Functions

Exponential & Log Equations; Semi-Log Plots

Core: Take logs to bring exponents down; check every log equation's domain.

Mon Nov 29
Same base
TRY Solve $5^{2x} = 125$.

Tue Nov 30
Using logs
TRY Solve $e^{0.2t} = 3$.

Wed Dec 1
Log equations
TRY Solve $\ln x + \ln(x - 1) = \ln 6$.

Thu Dec 2
Semi-log plots
TRY Data form a straight line on a semi-log plot. What kind of model fits?

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Trigonometric Functions

Periodic Phenomena & the Unit Circle

Core: On the unit circle, the point at angle θ is $(\cos\theta, \sin\theta)$.

Mon Dec 6
Periodic data
TRY A pattern repeats every 24 hours. What is its period?

Tue Dec 7
Radians
TRY Convert $\frac{7\pi}{6}$ to degrees.

Wed Dec 8
Unit circle values
TRY Find $\cos\frac{7\pi}{6}$ and $\sin\frac{7\pi}{6}$.

Thu Dec 9
Tangent
TRY Find $\tan\frac{3\pi}{4}$.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Trigonometric Functions

Sinusoidal Functions **Core:** For $y = a\sin(b(x-h)) + k$: amplitude $|a|$, period $\frac{2\pi}{|b|}$, midline $y = k$.

Mon Dec 13
Amplitude & midline
TRY Find the amplitude and midline of $y = -2\sin x + 3$.

Tue Dec 14
Period
TRY Find the period of $y = \cos\left(\frac{\pi x}{4}\right)$.

Wed Dec 15
Phase shift
TRY How is $y = \sin\left(2\left(x - \frac{\pi}{6}\right)\right)$ shifted from $y = \sin 2x$?

Thu Dec 16
Range
TRY Find the range of $y = 5\cos x - 1$.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Trigonometric Functions

Sinusoidal Models & the Tangent Function **Core:** Read the max, min and period from the situation, then build the model.

Mon Jan 3
Daylight
TRY Daylight ranges from 9 to 15 hours over a 365-day year. Find the amplitude and midline.

Tue Jan 4
Writing a model
TRY A quantity has a maximum of 10 at $t = 0$ and a minimum of 2 at $t = 6$. Write a cosine model.

Wed Jan 5
Tangent period
TRY Find the periods of $y = \tan x$ and $y = \tan 2x$.

Thu Jan 6
Tangent asymptotes
TRY Where does $y = \tan x$ have vertical asymptotes on $(0, 2\pi)$?

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Trigonometric Functions

Inverse Trigonometric Functions **Core:** Inverse trig functions return one angle from a restricted range.

Mon Jan 10
Inverse sine
TRY Find $\sin^{-1}\left(\frac{1}{2}\right)$.

Tue Jan 11
Inverse cosine
TRY Find $\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)$.

Wed Jan 12
Inverse tangent
TRY Find $\tan^{-1}(-1)$.

Thu Jan 13
Compositions
TRY Find $\sin\left(\cos^{-1}\frac{3}{5}\right)$.

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Trigonometric Functions

Mid-Year Review & Checkpoint **Core:** Review Units 1–4 and practice working steadily through a full test.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: logarithms
TRY Solve $\log_3 x = -2$.

Tue Jan 18
Review: exponential models
TRY A drug has a half-life of 6 hours. What fraction is left after 18 hours?

Wed Jan 19
Review: sinusoids
TRY Find the period of $y = 3\sin 4x$.

Thu Jan 20
Review: inverse trig
TRY Find $\cos^{-1} 0$.

Fri Jan 21
Mid-Year Checkpoint
Units 1–4 test. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Trig Identities, Polar & Complex Numbers

Trig Equations & Identities

Core: Isolate the trig function or factor, then use the unit circle.

Mon Jan 24
Solving
TRY Solve $2\sin x + \sqrt{3} = 0$ on $[0, 2\pi)$.

Tue Jan 25
Factoring
TRY Solve $\sin x \cos x - \sin x = 0$ on $[0, 2\pi)$.

Wed Jan 26
Pythagorean identities
TRY Simplify $1 + \tan^2 x$.

Thu Jan 27
Simplifying
TRY Simplify $\dfrac{1 - \sin^2 x}{\cos x}$.

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Trig Identities, Polar & Complex Numbers

Sum & Double-Angle Identities

Core: $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$; $\cos 2\theta = 1 - 2\sin^2 \theta$.

Mon Jan 31
Sum identities
TRY Find the exact value of $\cos 15°$.

Tue Feb 1
Difference identities
TRY Find $\sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$.

Wed Feb 2
Double angles
TRY If $\sin\theta = \frac{3}{5}$, find $\cos 2\theta$.

Thu Feb 3
Solving with double angles
TRY Solve $\sin 2x = \sin x$ on $[0, 2\pi)$.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Trig Identities, Polar & Complex Numbers

Secant, Cosecant & Cotangent

Core: $\sec x = \frac{1}{\cos x}$, $\csc x = \frac{1}{\sin x}$, $\cot x = \frac{\cos x}{\sin x}$.

Mon Feb 7
Values
TRY Find $\sec\frac{\pi}{3}$ and $\csc\frac{\pi}{6}$.

Tue Feb 8
Cotangent
TRY Find $\cot\frac{\pi}{4}$.

Wed Feb 9
Where undefined
TRY For which x is $\csc x$ undefined?

Thu Feb 10
Identity
TRY Simplify $1 + \cot^2 x$.

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Trig Identities, Polar & Complex Numbers

Polar Coordinates & Complex Numbers

Core: $x = r\cos\theta$, $y = r\sin\theta$, and $r^2 = x^2 + y^2$.

Mon Feb 14
Polar to rectangular
TRY Convert $\left(4, \frac{\pi}{3}\right)$ to rectangular coordinates.

Tue Feb 15
Rectangular to polar
TRY Convert $(-3, 3)$ to polar coordinates with $r > 0$.

Wed Feb 16
Complex numbers in polar form
TRY Write $1 + i$ in polar form.

Thu Feb 17
Polar equations
TRY Write $r = 4\cos\theta$ in rectangular form.

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Trig Identities, Polar & Complex Numbers

Polar Graphs **Core:** In a polar graph, r is the distance from the origin at angle θ.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Circles

TRY Describe the graph of $r = 3$.

Tue Feb 22

Roses

TRY How many petals does $r = 2\sin 3\theta$ have?

Wed Feb 23

Cardioids

TRY What is the greatest value of r for $r = 2 + 2\cos\theta$?

Thu Feb 24

Rate of change

TRY For $r = 2 + 2\cos\theta$, is r increasing or decreasing on $(0, \pi)$?

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 6: Parameters, Vectors & Matrices

Parametric Functions **Core:** A parameter t gives both $x(t)$ and $y(t)$, like position over time.

Mon Feb 28

Plotting

TRY For $x = t + 1$ and $y = t^2$, find the point at $t = 2$.

Tue Feb 29

Eliminating the parameter

TRY Eliminate t from $x = 2t$, $y = t + 3$.

Wed Mar 1

Circles

TRY Eliminate t from $x = 3\cos t$, $y = 3\sin t$.

Thu Mar 2

Motion

TRY A particle moves with $x = 4t$ and $y = 3t$. What is its speed?

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Parameters, Vectors & Matrices

Conic Sections **Core:** Parabolas, ellipses and hyperbolas are defined by distances to foci.

Mon Mar 6

Parabolas

TRY Find the vertex and focus of $y^2 = 8x$.

Tue Mar 7

Ellipses

TRY Find the foci of $\frac{x^2}{36} + \frac{y^2}{16} = 1$.

Wed Mar 8

Hyperbolas

TRY Find the asymptotes of $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

Thu Mar 9

Parametric conics

TRY Eliminate t from $x = 5\cos t$, $y = 2\sin t$.

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Parameters, Vectors & Matrices

Vectors **Core:** A vector has a size (magnitude) and a direction.

Mon Mar 13

Magnitude

TRY Find the magnitude of $\langle 6, -8\rangle$.

Tue Mar 14

Operations

TRY If $\mathbf{u} = \langle 2, 1\rangle$ and $\mathbf{v} = \langle -1, 3\rangle$, find $3\mathbf{u} - \mathbf{v}$.

Wed Mar 15

Unit vectors

TRY Find the unit vector in the direction of $\langle 3, 4\rangle$.

Thu Mar 16

Dot product

TRY Find $\langle 1, 2\rangle \cdot \langle 4, -2\rangle$. What does the result tell you?

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Parameters, Vectors & Matrices

Matrices **Core:** Multiply rows by columns; a 2×2 matrix has an inverse when its determinant is not 0.

Mon Mar 20
Adding & scaling
TRY Compute $2\begin{bmatrix}1 & 3\\ 0 & -1\end{bmatrix} + \begin{bmatrix}2 & 0\\ 1 & 1\end{bmatrix}$.

Tue Mar 21
Multiplying
TRY Compute $\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$.

Wed Mar 22
Determinants
TRY Find the determinant of $\begin{bmatrix}3 & 5\\ 1 & 2\end{bmatrix}$.

Thu Mar 23
Inverses
TRY Find the inverse of $\begin{bmatrix}2 & 1\\ 5 & 3\end{bmatrix}$.

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Parameters, Vectors & Matrices

Matrices as Transformations **Core:** A 2×2 matrix moves every point of the plane; its determinant scales area.

Mon Apr 3
Applying a matrix
TRY Apply $\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$ to $\langle 3, 1\rangle$. What transformation is it?

Tue Apr 4
Rotation matrices
TRY Write the matrix for a $180°$ rotation about the origin.

Wed Apr 5
Area scaling
TRY A transformation has determinant 3. What happens to a square of area 2?

Thu Apr 6
Composition
TRY What single transformation is a $90°$ rotation done twice?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Parameters, Vectors & Matrices

Matrix Models **Core:** A transition matrix moves a population from one step to the next.

Mon Apr 10
Transition matrices
TRY Each year 80% of city residents stay and 20% move to the suburbs; 10% of suburbanites move to the city. Start: 1,000 in the city, 0 in the suburbs. After 1 year?

Tue Apr 11
The next step
TRY Using the same model, find the populations after 2 years.

Wed Apr 12
Steady state
TRY In the long run, what fraction of the people live in the city?

Thu Apr 13
Systems with matrices
TRY Solve $\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} = \begin{bmatrix}5\\ 10\end{bmatrix}$.

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Year-End Review

Year-End Review: Functions & Models **Core:** Mix the skills from Units 1–3 and justify each answer.

Mon Apr 17
Review: rational functions
TRY Find the hole and the vertical asymptote of $f(x) = \dfrac{x^2 - x}{x^2 - 1}$.

Tue Apr 18
Review: exponential equations
TRY Solve $2^x = 10$.

Wed Apr 19
Review: logarithms
TRY Compute $\log_2 40 - \log_2 5$.

Thu Apr 20
Timed set: 10 in 20 min
TRY Find the average rate of change of $\sin x$ on $\left[0, \frac{\pi}{2}\right]$.

Fri Apr 21
Weekly Quiz
10 mixed review problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Year-End Review

Year-End Review & Practice Exam

Core: Practice a full exam, then learn from every error.

Mon Apr 24

Review: trig equations

TRY Solve $\cos 2x = 0$ on $[0, \pi)$.

Tue Apr 25

Review: polar coordinates

TRY Convert $(2, \pi)$ to rectangular coordinates.

Wed Apr 26

Review: vectors

TRY Find the magnitude of $\langle 5, 12 \rangle$.

Thu Apr 27

Practice Exam

Full-length AP-style practice exam covering the year.

Fri Apr 28

Error-log review

Redo every missed problem from the exam; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Year-End Review

Final Checkpoint & Celebration

Core: Show what you have learned and get ready for Calculus.

Mon May 1

Light review

TRY Find $\tan \frac{\pi}{3}$.

Tue May 2

Mixed warm-up set

TRY Find the determinant of $\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$.

Wed May 3

Final Checkpoint

Covers the whole year.

Thu May 4

Checkpoint review

Go over solutions together; students explain one problem each.

Fri May 5

Celebration & reflection

Certificates, progress review, and goals for Calculus next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 5
Tue: 20
Wed: 150 people per year
Thu: 1.5 and 6

Week 2
Mon: $x < 2$
Tue: Concave up
Wed: Concave down
Thu: At $x = 0$

Week 3
Mon: 0 (multiplicity 2), 3 (multiplicity 3)
Tue: 8
Wed: 3
Thu: 4

Week 4
Mon: $-\infty$
Tue: Even
Wed: $x < -2$ or $x > 1$
Thu: $x \le 0$ or $x = 3$

Week 5
Mon: $x = 1$ and $x = -3$
Tue: $y = \frac{2}{5}$
Wed: $y = x + 1$
Thu: ∞

Week 6
Mon: At $(2, 4)$
Tue: $x = 3$ ($x = -3$ is a hole)
Wed: $x < -1$ or $x \ge 2$
Thu: It grows to ∞ from both sides

Week 7
Mon: $x^2 + x + 1$
Tue: $x^4 + 8x^3 + 24x^2 + 32x + 16$
Wed: 3
Thu: $x = \pm 2, \pm 3$

Week 8
Mon: Vertical stretch by 2, right 3, up 1
Tue: Horizontal compression by a factor of $\frac{1}{2}$
Wed: $[-5, 10]$
Thu: $(-4, 2)$

Week 9
Mon: Quadratic
Tue: \$11
Wed: $y = 2x^2 + x + 3$
Thu: Growth usually is not constant

Week 10
Mon: 3
Tue: 0 and 2 (double)
Wed: $x = -4$ and $y = 3$
Thu: $(-1, 4)$

Week 11
Mon: -20
Tue: 384
Wed: Exponential
Thu: 2

Week 12
Mon: $2^{1/5} \approx 1.149$
Tue: $y = 800(0.88)^t$
Wed: $12 \cdot 4^x$
Thu: About 134.99

Week 13
Mon: 4
Tue: $f(x) = x^5$, $g(x) = 3x - 1$
Wed: $f^{-1}(x) = \log_2(x + 1)$
Thu: $x > 2$

Week 14
Mon: -4
Tue: 2
Wed: $2 \ln x + \frac{1}{2} \ln y$
Thu: $x = -3$

Week 15
Mon: $x = \frac{3}{2}$
Tue: $t = 5 \ln 3 \approx 5.49$
Wed: $x = 3$
Thu: Exponential

Week 16
Mon: 24 hours
Tue: $210°$
Wed: $-\frac{\sqrt{3}}{2}$ and $-\frac{1}{2}$
Thu: -1

Week 17
Mon: Amplitude 2; midline $y = 3$
Tue: 8
Wed: Right $\frac{\pi}{6}$
Thu: $[-6, 4]$

Week 18
Mon: Amplitude 3 h; midline 12 h
Tue: $y = 4\cos\left(\frac{\pi t}{6}\right) + 6$
Wed: π and $\frac{\pi}{2}$
Thu: $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$

Week 19
Mon: $\frac{\pi}{6}$
Tue: $\frac{3\pi}{4}$
Wed: $-\frac{\pi}{4}$
Thu: $\frac{4}{5}$

Week 20
Mon: $x = \frac{1}{9}$
Tue: $\frac{1}{8}$
Wed: $\frac{\pi}{2}$
Thu: $\frac{\pi}{2}$

Week 21
Mon: $\frac{4\pi}{3}$, $\frac{5\pi}{3}$
Tue: $0, \pi$
Wed: $\sec^2 x$
Thu: $\cos x$

Week 22
Mon: $\frac{\sqrt{6}+\sqrt{2}}{4}$
Tue: $\frac{\sqrt{6}-\sqrt{2}}{4}$
Wed: $\frac{7}{25}$
Thu: 0, $\frac{\pi}{3}$, π, $\frac{5\pi}{3}$

Week 23
Mon: 2 and 2
Tue: 1
Wed: $x = k\pi$ for any integer k
Thu: $\csc^2 x$

Week 24
Mon: $(2, 2\sqrt{3})$
Tue: $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$
Wed: $\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$
Thu: $x^2 + y^2 = 4x$

Week 25
Mon: A circle of radius 3 centered at the origin
Tue: 3
Wed: 4
Thu: Decreasing

Week 26
Mon: $(3, 4)$
Tue: $y = \frac{x}{2} + 3$
Wed: $x^2 + y^2 = 9$
Thu: 5 units per unit of time

Week 27
Mon: Vertex $(0, 0)$; focus $(2, 0)$
Tue: $(\pm 2\sqrt{5}, 0)$
Wed: $y = \pm\frac{4}{3}x$
Thu: $\frac{x^2}{25} + \frac{y^2}{4} = 1$

Week 28
Mon: 10
Tue: $\langle 7, 0\rangle$
Wed: $\langle \frac{3}{5}, \frac{4}{5}\rangle$
Thu: 0: the vectors are perpendicular

Week 29
Mon: $\begin{bmatrix} 4 & 6 \\ 1 & -1 \end{bmatrix}$
Tue: $\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$
Wed: 1
Thu: $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$

Week 30
Mon: $\langle -1, 3\rangle$; a $90°$ rotation
Tue: $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$
Wed: Its image has area 6
Thu: A $180°$ rotation

Week 31
Mon: City 800, suburbs 200
Tue: City 660, suburbs 340
Wed: $\frac{1}{3}$
Thu: $x = 1, y = 3$

Week 32
Mon: Hole at $\left(1, \frac{1}{2}\right)$; asymptote $x = -1$
Tue: $x = \log_2 10 \approx 3.32$
Wed: 3
Thu: $\frac{2}{\pi}$

Week 33
Mon: $\frac{\pi}{4}$, $\frac{3\pi}{4}$
Tue: $(-2, 0)$
Wed: 13

Week 34
Mon: $\sqrt{3}$
Tue: 10

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